

Agents, Tools and Function Calling Assignments

ASSIGNMENT 1 – ReAct Agent – multi step reasoning

Modify the **Weather Agent** you created so it can answer the prompt:

“Should I carry an umbrella tomorrow”

1. You must use the LLM as the planning, reasoning and action (ReAct) agent for the above.
2. If the location is not mentioned in the prompt, the chatbot should prompt the user to specify the location
3. The response from the LLM should mention the location and the reasoning for the advise it is giving. For example, the response should say:
“You do not need to carry umbrella tomorrow as the weather in New York is sunny”
4. Demonstrate using streamlit UI
5. Push your application to HuggingFace (HF)

Submission:

huggingFace URL

ASSIGNMENT 2 – News summarizer Agent

Create an app with streamlit that will

A) take any topic or company name or string/keywords as input from the user

B) Print **top 5** latest news on the topic. Each news item should print:

- i) Headline (relevant to the topic – may not be same as the article/news headline)
- ii) 3-5 line summary of each news item

Use <https://newsapi.org/> and <https://serper.dev/> (Google news) to fetch the news items

Use Agentic AI to build it.

Push the app to HF

Submit:

HF URL for your app.

ASSIGNMENT 3 – Stock recommender Agentic AI with crew.ai

Implements an AI-powered stock recommendation system using CrewAI. The system analyzes financial data for a specified company or stock, evaluates its performance and market trends, and provides investment recommendations. It's designed to help investors make informed decisions by leveraging multiple specialized AI agents that work together to gather information, analyze data, and generate actionable insights.

Use Amazon Bedrock LLM and Crew Workers to build the workflow

Submit: Colab notebook URL

ASSIGNMENT 4 – Financial planner Agentic AI with LangGraph

Build a ****simple multi-agent finance system**** using LangGraph.

It should use both ****normal flow**** and ****conditional logic**** to demonstrate how LangGraph helps you create smart agent workflows.

Use Amazon Bedrock LLM and Langchain Tools for the same

Use the provided starter code

Submit: Colab notebook URL